Instrumentation & Measurement

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Instrumentation & Measurement (A05921)

Short Title:	Instrumentation & Measurement		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Engineering		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Introductory

Description of Module / Aims

The module is designed to give a student the ability to understand, build and perform basic calculations on practical analog and digital electronic circuits commonly used in electrical engineering. Sensors commonly used in electrical engineering applications are covered. Linear operational amplifiers circuits such as basic amplifiers, instrumentation, signal conditioning circuits and interface circuits and essential non linear circuits like comparators and Schmitt triggers are included. It also covers ADC conversion and common 555 timer applications. The digital element concentrates on counters and registers. The essential differences between different logic families will be included. The module has a definite emphasis on industrial standards and typical industrial equipment.

Programmes

	BEng (Hons) in Automation Engineering with Data Intelligence (`WD_EAUTO_B`)	2 / 3 / M
	BEng (Hons) in Electrical Engineering (`WD_ETRIC_B`)	2 / 3 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	2 / 3 / M
	BEng (Hons) (`WD_ETRIC_B`)	2 / 3 / M
`ENGR-0087`	BEng in Electrical Engineering (`WD_EELCT_D`)	2 / 3 / M
	BSc (Hons) in the Internet of Things (International) (`WD_KINTT_BI`)	2 / 3 / M
`ENGR-0087`	Higher Certificate in Engineering in Electrical Engineering (`WD_EELCT_C`)	2 / 3 / M

Indicative Content

- Sensors (pressure, flow rate, level, humidity, ph, temperature, speed, position etc).
- Operational amplifier characteristics (voltage range, current drive, bandwidth, input and output impedance) and negative feedback
- Practical linear and non linear operational amplifier circuits (amplifier's, summers, instrumentation amplifiers, integrators, differentiators, comparators, Schmitt triggers)
- 555 timer and applications
- Flip flops, counters, registers and logic families
- Opto-Electronics
- Interface circuits and ADC conversion

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe typical sensor operation, calibration and usage (pressure, flow rate, level, temperature, speed, position etc).
2. Define the operation of fundamental linear and non linear operational amplifier circuits.
3. Construct and test fundamental linear and non linear operational amplifier circuits.
4. Explain typical instrumentation techniques and ADC conversion.
5. Describe and use common 555 timer based circuits.
6. Explain the operation of flip flops, counters and registers.

Learning and Teaching Methods

- Two hour practical laboratory sessions to build, test and document circuits
- Lab project based on circuits covered in lectures and labs but used to implement an electronics based solution to an instrumentation and measurement problem
- Lectures and integrated tutorials using typical classroom resources and Moodle and other on-line resources.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,4,5,6
Continuous Assessment	40%	
Practical	40%	3,5,6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Alciatore, D. _Introduction to Mechatronics & Measurement Systems_. 4th ed.. USA: McGraw Hill, 2011.
- Malvino, A. _Electronic Principles_. 8th ed.. USA: McGraw Hill, 2015.
- Nahvi, M. and J. Edminister. _Electric Circuits_. 4th ed.. USA: Schaum Outline Series, 2002.

Supplementary Material

- Tocci, R.J. and N. Widner. _Digital Systems: Principles and applications_. 12th. USA: Pearson, 2016.

Requested Resources

- ENGINEERING LAB: Electronics